Midterm Review

Katie Markovich, EPS 522/ ENVS 423L (Fall 2025)

Housekeeping

- PS #5 assigned soon, but not due until October 16
- Streamflow concepts will not be on Midterm, but will be on Final
- Midterm exam next Tuesday
- Please set a reminder to bring a calculator!
- I will have a handful of calculators available

Grading Breakdown

10 problem sets total, with one automatic 100% and one problem set that will be a peer evaluation of the 522 term project presentations. Four down and four to go!

ENVS 423L	
Homeworks	50%
Midterm	25%
Final	25%

EPS 522	
Homeworks	**30%**
Midterm	20%
Final	20%
Term Project	**30%**

Exam Details

- Six short answer question topics
- Several are lifted directly from problem sets
- Several are slight modifications from problem sets
- One involves writing out definitions of key terms

Topics Covered

- Hydrologic Concepts
- Climate System
- Precipitation and Snowmelt Processes
- Evapotranspiration

Hydrologic Concepts

- Regional water balance
- Basic unit conversions (mm/cm/m and cm/in)
- Primary dimensions are:
- Review dimensional analysis example in lecture slides

Climate System

- Basics of solar radiation
- Global energy balance
- Factors affecting variation of TOA solar radiation
- Major aspects of general circulation (and how it controls climate)

Precipitation and Snowmelt Processes

- Requirements for precipitation to occur
- Uplift mechanisms
- Measuring rain (and associated challenges)
- Areal estimation techniques
- Rainfall statistics

Evapotranspiration

- Evaporation types and processes
- Methods for estimating potential evaporation
- Transpiration process and estimation methods
- Interception loss

"Cheat Sheet"

You are allowed to bring an 8.5x11" sheet of white printer paper with handwritten notes, equations, etc. on one side.

You will need to turn this in with the exam, so please write your name at the top of it.

Questions?

Job Opportunities

Marcy Litvak is hiring the following:

1. Full-time tower technician (1-year, up to 3-years)
2. Data manager to process flux tower data
3. Technician
4. Student
5. Postdoc working on scaling fluxes to the watershed scale using machine learning

Please reach out to her for more information: mlitvak@unm.edu